STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-COMPLETION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1308-1318 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 1320-1339, the complete author proof (18 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 29 of queue_examples-defos.json (pb=1020, stmt 1308-1318, proof 1320-1339).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 05GH: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1339–1339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00MB: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1339–1339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00MA: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1339–1339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DY1: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1031–1040. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Example 0DY1 let P be a k-algebra. Assume P Noetherian and let $J \subset P$ be an ideal. Denote $P^\wedge$ the J-adic completion. There is a natural functor*

$$\mathcal{D}ef_P \longrightarrow \mathcal{D}ef_{P^\wedge}$$

of deformation categories.

Date: stacks-project:examples-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Given a deformation of P we can take the completion of it to get a deformation of the completion; this is clear and we encourage the reader to skip the proof. More precisely, let $(A, Q) \to (k, P)$ be a morphism in $\mathcal{F}$, i.e., an object of $\mathcal{D}ef_P$. Observe that Q is a Noetherian ring: the kernel of the surjective ring map $Q \to P$ is nilpotent and finitely generated and P is Noetherian; apply Algebra, Lemma 05GH. Denote $J_Q \subset Q$ the inverse image of J in Q. Let $Q^\wedge$ be the J_Q-adic completion of Q. Recall that $Q \to Q^\wedge$ is flat (Algebra, Lemma 00MB) and hence $Q^\wedge$ is flat over A. The induced map $Q^\wedge \to P^\wedge$ induces an isomorphism $Q^\wedge \otimes_A k = Q^\wedge \otimes_Q P = P^\wedge$ by Algebra, Lemma 00MA for example. Hence $(A, Q^\wedge) \to (k, P^\wedge)$ is the desired object of $\mathcal{D}ef_{P^\wedge}$. □